

Verification and Validation of the DataCenter Gas-Plant and Turbine-Governor Models

Equations, methods, and results — written to be readable by a newcomer to power-system dynamics

DataCenter repository V&V campaign

July 18, 2026

Abstract

This report documents the models implemented in the **DataCenter** repository — the IEEE PES-TR1 GGOV1 and TGOV1 turbine-governor models, a two-mass gas-path model of the GE LM2500 aeroderivative gas turbine, a torsional shaft-fatigue model, and the steady-state gas-plant thermodynamic surrogates — together with the full verification and validation (V&V) campaign performed on them. Every equation solved by the code is derived here from first principles, with per-unit conventions spelled out, because unit-system mistakes were the single largest class of defect found in the pre-V&V code. Headline results: the swing-equation + TGOV1 implementation matches the independent ANDES simulator to 0.124 mHz; the GGOV1 implementation reproduces the Hannett & Khan (1993) literature benchmark step response to 1.3%; and a Rowen-model reconstruction of the published Beluga 5 load-rejection event matches the recorded speed trace with an RMS misfit of 1.1×10^{-3} pu. All 36 automated tests pass. A companion running log (`docs/vv_log.md`) records every change, and a constants dossier (`docs/constants.md`) classifies every number in the models as pinned, estimated, or placeholder.

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1 Introduction

1.1 What system are we modeling?

The engineering question behind this repository: *can a gas-turbine power plant, disconnected from the bulk electric grid (“islanded”), keep an AI data center alive?* AI training workloads swing their electrical demand by large fractions in seconds. On the big interconnected grid those swings are absorbed by thousands of machines; on an island with one or a few gas turbines, every swing is a direct fight between the load and the turbine’s control system. The models here simulate that fight.

Three physical subsystems interact:

1. **The rotating machine.** A gas turbine spins a synchronous generator. The rotational kinetic energy stored in the shaft is the system’s first line of defense: when load suddenly exceeds mechanical power, the shaft slows down and that kinetic energy covers the deficit for the first second or so. Shaft speed *is* grid frequency, so “the shaft slows down” means “the lights flicker at 59.5 Hz.”
2. **The governor.** A feedback controller that measures shaft speed and moves the fuel valve to restore it. The industry describes governors with standardized block diagrams — TGOV1 (simple) and GGOV1 (detailed, gas-turbine-specific) from the IEEE PES Task Force report PES-TR1 (2013) [1]. Using the standard models is what makes results communicable to grid engineers.
3. **The thermodynamic plant.** Fuel in, electricity and hot exhaust out. Steady-state surrogate models convert operating points into fuel burn, efficiency, exhaust conditions, and CO₂.

1.2 What do “verification” and “validation” mean?

For the newcomer: V vs. V

Verification asks: *did we solve the equations right?* It compares the code against mathematics — analytic solutions, invariants that must hold exactly (energy balances, steady-state identities), and independent implementations of the same equations (here, the ANDES simulator). **Validation** asks: *did we solve the right equations?* It compares the model against reality — here, published field tests on real gas turbines (Hannett & Khan 1993) and published thermodynamic data. A model can pass verification and fail validation (a perfectly-solved wrong model) and vice versa.

This campaign ran in phases: Phase 0 fixed defects found in a code review (they are cataloged in Appendix A — read it; several are instructive); Phase 1 verified; Phase 2 audited every constant; Phase 3 validated against literature; Phase 4 froze everything into an automated regression suite (`pixi run vv-all`).

2 Foundations: per-unit systems and the swing equation

2.1 The per-unit system

Power-system engineers divide every quantity by a *base* value and work with the dimensionless ratios, called *per-unit* (pu) values.

For the newcomer: Why per-unit?

If a generator is rated 23 MVA and currently produces 15 MW, its output is $15/23 = 0.652$ pu. The pu system makes machines of different sizes directly comparable (“the valve is at 65 % of rating”) and makes the equations cleaner. The catch — and the source of the worst bug found in this code — is that *a pu number is meaningless until you know its base*. The same physical 22 MW is 0.957 pu on a 23 MVA generator base but 1.000 pu on a 22 MW turbine base.

A worked example. The LM2500 gen-set: $S_n = 23$ MVA (generator rating), $T_{\text{rate}} = 22$ MW (turbine rating). The unit carries 15 MW:

$$P_e^{(g)} = \frac{15}{23} = 0.652 \text{ pu (gen base)}, \quad P_e^{(t)} = \frac{15}{22} = 0.682 \text{ pu (turbine base)}.$$

The governor needs the turbine-base number (its fuel map is defined against the turbine rating); the swing equation needs the gen-base number (its inertia constant is defined against the MVA rating). The fuel valve sits at $v = P_e^{(t)}/K_{\text{turb}} + W_{\text{fnl}} = 0.682/1.5 + 0.2 = 0.655$ pu stroke — note this is a *third* normalized quantity (fuel-flow units), numerically close to the power numbers only by coincidence of K_{turb} and W_{fnl} . Defect G2 (Appendix A) happened precisely because a fuel-stroke number was passed somewhere a power fraction was expected.

Two bases coexist in this repository, deliberately:

- **Generator base** $S_n = 23$ MVA: used by the swing equation (rotor mechanics), because machine inertia data is quoted on it.
- **Turbine base** $T_{\text{rate}} = 22$ MW: used by *everything inside the governor* (valve stroke, fuel flow, mechanical power, limits), because IEEE PES-TR1 §3.3 defines the GGOV1 parameters on it (PSS/E calls this base **Trate**).

The single conversion constant is

$$k_b = \frac{T_{\text{rate}}}{S_n} = \frac{22}{23} = 0.9565, \quad P^{(\text{gen base})} = k_b P^{(\text{turbine base})}. \quad (1)$$

It is applied at exactly two places: mechanical power entering the swing equation, and electrical power entering the governor’s power transducer. Appendix A (defect G1) describes what went wrong when the pre-V&V code tried to blend the two bases by hand-editing one limit.

2.2 The swing equation

Newton’s second law for a rotating shaft with moment of inertia J (kg m^2), mechanical drive torque τ_m and electrical braking torque τ_e (Nm):

$$J \frac{d\Omega}{dt} = \tau_m - \tau_e, \quad (2)$$

where Ω is the mechanical angular speed (rad/s). Define the *inertia constant*

$$H = \frac{\frac{1}{2} J \Omega_0^2}{S_n} \quad (\text{units: seconds}), \quad (3)$$

the kinetic energy at rated speed divided by the machine rating. H answers: *for how many seconds could the stored kinetic energy alone supply rated power?* Aeroderivative gen-sets are light: $H \approx 1.5\text{--}3.5$ s. Normalizing speed as $\omega = \Omega/\Omega_0$ and power by S_n , and using $P = \tau \Omega$, the equation becomes (to first order near $\omega = 1$) the standard *swing equation*:

$$\boxed{2H \frac{d\omega}{dt} = P_m - P_e - D(\omega - 1)} \quad \frac{d\delta}{dt} = \omega_0(\omega - 1), \quad (4)$$

with $\omega_0 = 2\pi \cdot 60$ rad/s, rotor electrical angle δ , and damping coefficient D . This “power form” ($P_m - P_e$ rather than the strict torque form $(P_m - P_e)/\omega$) was verified in Phase 1 to be *exactly* what the ANDES simulator solves: ANDES’s classical machine uses the standard speed-voltage approximation (stator flux evaluated at $\omega = 1$), under which electrical torque equals electrical power, and its governor output enters the torque equation one-to-one (§9.1).

For the newcomer: Frequency, speed, and why 60 Hz

The generator here has 2 magnetic poles, so electrical frequency equals shaft speed: 3600 rpm = 60 rev/s = 60 Hz. When $\omega = 0.99$ pu, the grid is at 59.4 Hz. Data-center power supplies and the turbine itself have under/over-frequency protection, so excursions of a few hundred mHz matter and excursions of a few Hz trip equipment.

2.3 Load damping

Real loads draw slightly less power when frequency drops (motors slow down, etc.). This is modeled on the load, not the machine:

$$P_e(t, \omega) = P_{\text{demand}}(t) [1 + \alpha(\omega - 1)], \quad \alpha = 1.5, \quad (5)$$

with α in Kundur’s typical range 1–2 [4]. Because the load damping is explicit, the machine damping is set to $D = 0$: the pre-V&V code carried *both* $D = 1$ and $\alpha = 1.5$, double counting the same physics and making every predicted frequency dip optimistic (defect G5).

2.4 Droop: how multiple governors share load

A governor could hold frequency exactly at 60 Hz (*isochronous* control), but two isochronous governors on one grid fight each other. The standard solution is *droop*: the governor deliberately lets steady-state frequency fall linearly with its power output,

$$\Delta\omega_{ss} = -R \Delta P_{ss}, \quad R = 0.04 \text{ (4\%)}, \quad (6)$$

i.e., going from no load to full load lowers the machine’s steady frequency target by 4% (2.4 Hz), so machines share load in proportion to their ratings without communicating. For a *single* islanded unit, isochronous control (or droop plus a slow secondary loop) is standard practice; the implementation supports both (`rselect` parameter, §4.3), and the droop identity (6) is one of the exact tests in the verification suite.

3 The TGOV1 governor model (legacy, Tier A)

TGOV1 is the simplest standard turbine-governor model: one lag, one lead-lag, one droop gain. In this project it is the *legacy* model (kept because it is what ANDES can cross-simulate and it anchors

the project’s history); GGOV1 is the reference model. The block structure, now implemented exactly per the standard:

$$\underbrace{\frac{1}{R}(\Delta\omega_{\text{ref}} - \Delta\omega) + P_{\text{ref}}}_{\text{droop input}} \longrightarrow \underbrace{\frac{1}{1+sT_1}}_{\substack{\text{valve servo,} \\ \text{non-windup } [V^{\min}, V^{\max}]}} \longrightarrow \underbrace{\frac{1+sT_2}{1+sT_3}}_{\text{turbine lead-lag}} \longrightarrow P_m - D_t \Delta\omega. \quad (7)$$

For the newcomer: Reading transfer-function blocks

$\frac{1}{1+sT}$ is a *first-order lag*: the output chases the input with time constant T (63 % of a step change after T seconds). In state-space form, $\dot{x} = (u - x)/T$. The lead-lag $\frac{1+sT_2}{1+sT_3}$ responds immediately with a fraction T_2/T_3 of a step and then relaxes to the full value with time constant T_3 — it models the fact that some turbine power responds quickly to the valve and the rest waits for the gas path to fill.

The ODE system solved (state vector $[\delta, \omega, v, x_2, P_{\text{fuel}}]$, all pu on the *generator* base — valid for TGOV1 because its implicit turbine gain is 1, so valve pu = power pu):

$$\dot{v} = \frac{1}{T_1} \left[P_{\text{ref}} + \frac{1-\omega}{R} - v \right] \quad (\text{zeroed if it would push } v \text{ past } V^{\min}/V^{\max}), \quad (8)$$

$$\dot{x}_2 = \frac{v - x_2}{T_3}, \quad P_m = x_2 + \frac{T_2}{T_3}(v - x_2) - D_t(\omega - 1), \quad (9)$$

$$2H\dot{\omega} = P_m - P_e(t, \omega), \quad \dot{P}_{\text{fuel}} = \frac{v - P_{\text{fuel}}}{T_{\text{comb}}}. \quad (10)$$

Parameters: $R = 0.04$, $T_1 = 0.15$ s, $T_2 = 0.3$ s, $T_3 = 1.5$ s, $V^{\max} = 22/23$ (the turbine rating on the generator base — correct *here* because valve pu = power pu), $V^{\min} = 0.15$.

V&V finding: slew limits inside a tight loop cause limit cycles

The pre-V&V code added a ± 0.1 pu/s valve rate limit. Placed at the *standard* limiter location (on the valve state, inside the droop loop), this produces a sustained ± 1.3 Hz oscillation at LM2500 gains: the loop gain is $1/(RT_1) \approx 167 \text{ s}^{-1}$, the commanded slew hugely exceeds the limit during transients, and the resulting phase lag destabilizes the loop — a classic describing-function instability, not a numerical artifact. (The old code “worked” only because its rate limiter sat outside the feedback path, a nonstandard structure that also added a spurious second pole at $1/T_1$.) Standard TGOV1 has no rate limit; rate-limited fuel dynamics belong in GGOV1, whose ± 0.05 error clamp bounds the commanded rate. The option is now off by default.

4 The GGOV1 governor model (reference, Tier B)

GGOV1 (PES-TR1 Fig. 3-5) is the industry-standard gas-turbine governor model. It captures the three control loops every modern GT control system (GE Speedtronic, Woodward) actually runs, arbitrated by a *low value select* (LVS): whichever controller asks for the *least* fuel wins.

4.1 State vector and parameters

The implementation integrates twelve states:

state	meaning	base	equation
δ	rotor electrical angle (rad)	—	$\dot{\delta} = \omega_0(\omega - 1)$
ω	rotor speed	pu of $2\pi 60$	swing eq. (4)
$\tilde{P}_e$	measured electrical power	turbine	T_{pelec} lag
x_i	speed-PI integrator	turbine	(17)
x_{ka}	acceleration integrator ($= f_{sra}$)	turbine	(17)
x_a	speed lag for the $d\omega/dt$ filter	pu	T_a lag
x_{iT}	temperature-PI integrator	turbine	(17)
x_{tload}	temperature-signal lag	turbine	T_{fload} lag
x_{tsab}	temperature-signal lead-lag state	turbine	T_{sa}/T_{sb}
v	fuel-valve stroke	turbine	actuator + limits
x_t	turbine lead-lag state	turbine	T_b/T_c on W_f
P_{fuel}	combustor-lagged fuel signal	turbine	T_{comb} lag

Table 1: GGOV1 state vector as implemented (`gas_plant/dynamics/ggov1.py`).

param	value	unit	status	param	value	unit	status
R	0.04	pu	P	K_{turb}	1.5	—	P
T_{pelec}	1.0	s	P	W_{fnl}	0.2	pu	P
Max/MinERR	± 0.05	pu	P	T_b, T_c	0.1, 0	s	P
K_{pgov}	10	—	P*	V^{max}, V^{min}	1.0, 0.15	pu	P
K_{igov}	2	—	P*	$R_{open/close}$	± 0.10	pu/s	P
a_{set}	0.01	pu/s	P	L_{dref}	1.0	pu(t)	P
K_a, T_a	10, 0.1	—, s	P	K_{pload}, K_{iload}	2, 0.67	—	P
T_{act}	0.15	s	E	$T_{fload}, T_{sa}, T_{sb}$	3, 4, 5	s	P
H	2.8	s	PL	D_m	0	—	P
α	1.5	—	P(range)	η_{design}	0.365	—	E

Table 2: Parameter values with dossier status: P = pinned (PES-TR1 Appendix C unless noted), E = estimated, PL = placeholder. *pinned to the *generic* PES-TR1 value; flagged for vendor retune (Hannett’s field-derived governors are 2–4 $\times$ hotter). Full provenance per row in `docs/constants.md`.

4.2 Per-unit convention

All governor signals below — valve stroke v , fuel flow W_f , the controller outputs $f_{srn}, f_{sra}, f_{srt}$, mechanical power $P_m^{(t)}$, and the limits — are pu on the *turbine base* $T_{rate} = 22$ MW. The swing equation stays on the generator base; the conversion (1) appears exactly twice:

$$2H \dot{\omega} = k_b P_m^{(t)} - P_e^{(g)} - D(\omega - 1), \quad P_e^{(t)} = P_e^{(g)} / k_b. \quad (11)$$

4.3 Loop 1: speed/power governor

A PI controller on a speed error that includes selectable droop feedback:

$$e_s = \begin{cases} (1 - \omega) + R(P_{ref} - \tilde{P}_e) & \text{rselect} = 1 \text{ (droop on electrical power; default)} \\ (1 - \omega) & \text{rselect} = 0 \text{ (isochronous)} \\ (1 - \omega) + R(P_{ref} - v) & \text{rselect} = -1 \text{ (droop on valve stroke)} \\ \frac{(1 - \omega) + R(P_{ref} - x_i)}{1 + R K_p} & \text{rselect} = -2 \text{ (droop on governor output)} \end{cases} \quad (12)$$

clamped to $[\text{MinERR}, \text{MaxERR}] = [-0.05, +0.05]$, then

$$f_{srn} = K_{pgov} e_s + x_i, \quad \dot{x}_i = K_{igov} e_s - (\text{anti-windup, §4.6}). \quad (13)$$

Here $\tilde{P}_e$ is the electrical power measured through a transducer lag $\dot{\tilde{P}}_e = (P_e^{(t)} - \tilde{P}_e)/T_{pelec}$. The `rselect` = -2 case contains an algebraic loop (e_s depends on f_{srn} which depends on e_s); it is solved in closed form as shown. The pre-V&V code documented four `rselect` modes but implemented only one — notably missing isochronous, the natural mode for a single islanded unit (defect G4).

4.4 Loop 2: acceleration limiter

The rotor acceleration is estimated with a washout filter (the exact state-space realization of $s/(1 + sT_a)$):

$$\dot{x}_a = \frac{\omega - x_a}{T_a}, \quad \hat{a} = \frac{\omega - x_a}{T_a}, \quad e_a = a_{\text{set}} - \hat{a}, \quad f_{sra} = x_{ka}, \quad \dot{x}_{ka} = K_a e_a - (\text{a.w.}) \quad (14)$$

It only takes over when the shaft accelerates faster than $a_{\text{set}} = 0.01$ pu/s (startup, load rejection).

4.5 Loop 3: temperature (thermal) limiter

Gas turbines are power-limited by hot-section temperature. GGOV1 proxies exhaust temperature by fuel flow: W_f passes through a lead-lag $(1 + sT_{sa})/(1 + sT_{sb})$ (thermocouple/radiation-shield dynamics) and a lag $1/(1 + sT_{fload})$, giving x_{tload} ; the limiter then runs a PI on

$$e_T = \underbrace{\left(\frac{L_{dref}}{K_{turb}} + W_{fnl} \right)}_{t_{lim} = \text{fuel flow at the thermal limit}} - x_{tload}, \quad f_{srt} = K_{pload} e_T + x_{iT}, \quad \dot{x}_{iT} = K_{iload} e_T - (\text{a.w.}) \quad (15)$$

With $L_{dref} = 1.0$ on the turbine base, the limiter caps continuous output at exactly $1.0 \text{ pu} \times 22 \text{ MW} = 22 \text{ MW}$ — this is *the* enforcement of the turbine's thermal rating (verified by test: a 25 MW demand settles at $P_m = 22.000 \text{ MW}$ with the temperature limiter selected). The pre-V&V code, with its mixed bases, enforced 23 MW here and 26.1 MW at the valve limit while claiming 22 MW (defect G1).

4.6 Low value select and anti-windup

$$f_{sr} = \min(f_{srn}, f_{sra}, f_{srt}). \quad (16)$$

For the newcomer: Integrator windup

A PI controller's integrator keeps accumulating error even while its output is being ignored (not selected by the LVS, or the valve is pinned at a limit). When conditions change, the wound-up integrator forces a long, sluggish overshoot before the controller re-engages. Every practical multi-loop controller needs *anti-windup*.

This implementation uses an always-active *back-calculation* form on all three integrators:

$$\dot{x}_i = K_i e - K_{bc} (f_{srX} - f_{sr}). \quad (17)$$

Because f_{sr} is the minimum, $(f_{srX} - f_{sr}) \geq 0$ always: it is *zero* for the selected controller (pure PI, no distortion) and positive for unselected ones, pulling their integrators down to track the actual

command so they re-engage cleanly. The form is continuous (no if/else switching), which keeps the ODE solver's step size healthy. The tracking gain K_{bc} is an implementation knob, not physics; Phase 1 verified that varying it over a $25\times$ range moves a step-response nadir by only 2.1 mHz.

4.7 Valve actuator and turbine

$$\dot{v} = \text{clip}\left(\frac{\text{clip}(f_{sr}, V^{min}, V^{max}) - v}{T_{act}}, R_{close}, R_{open}\right) \quad (\text{zeroed at the position limits}), \quad (18)$$

$$W_f = v \cdot \omega^{[\text{flag}=1]} \cdot \omega^{D_m[D_m < 0]}, \quad \dot{x}_t = \frac{W_f - x_t}{T_b}, \quad (19)$$

$$P_m^{(t)} = K_{turb}(x_t + T_c \dot{x}_t - W_{fnl}) - D_m(\omega - 1)[D_m > 0]. \quad (20)$$

Two constants deserve explanation for the newcomer:

- $W_{fnl} = 0.2$ is the *full-speed no-load* fuel: a spinning, unloaded gas turbine still burns $\sim 20\%$ of rated fuel just to drive its own compressor. Power is proportional to fuel *above* that floor — hence $P_m = K_{turb}(W_f - W_{fnl})$ with $K_{turb} = 1.5 \approx 1/(1 - W_{fnl} \dots)$ chosen so $W_f = 0.867$ gives rated power.
- $V^{max} = 1.0$ is a valve *stroke* limit, not a power limit. At the stroke limit the turbine could transiently make $K_{turb}(1.0 - 0.2) = 1.2 \text{ pu} = 26.4 \text{ MW}$ — short-term capability that the temperature limiter then winds back to 22 MW over its T_{fload}/PI time scale. Both ceilings are now explicit, tested properties.
- D_m : for $D_m > 0$ the term *subtracts* power as speed rises (damping). The pre-V&V code had the sign flipped — energy injection on overspeed (defect G3); the test suite now pins the direction (a load rejection peaks at 75.1 Hz with $D_m=0$ vs 72.0 Hz with $D_m=0.5$).

4.8 Steady-state initialization

Transient studies must start from an exact equilibrium or the first seconds of every run are polluted by artificial settling. Setting all derivatives to zero at $\omega = 1$, load $P_{e0}^{(t)}$:

$$v_0 = \frac{P_{e0}^{(t)}}{K_{turb}} + W_{fnl}, \quad x_{i,0} = f_{sr,0} = v_0, \quad x_{ka,0} = v_0 + \frac{K_a a_{\text{set}}}{K_{bc,a}}, \quad x_{iT,0} = v_0 + \frac{K_{iload} e_{T0}}{K_{bc,T}} - K_{pload} e_{T0}, \quad (21)$$

where the small offsets come from balancing the back-calculation terms (17) of the *unselected* loops. Phase 1 verified $\|\text{RHS}\| < 3 \times 10^{-15}$ at five operating points — the initial condition is an equilibrium to machine precision.

4.9 Fuel accounting

Fuel mass flow is computed *natively* from the governor's own fuel signal:

$$\dot{P}_{\text{fuel}} = \frac{W_f - P_{\text{fuel}}}{T_{\text{comb}}}, \quad \dot{m}_{\text{fuel}} = w_{\text{base}} P_{\text{fuel}}, \quad w_{\text{base}} = \frac{T_{\text{rate}}}{\eta_{\text{design}} \text{LHV} \left(\frac{L_{\text{dref}}}{K_{turb}} + W_{fnl} \right)} = 1.419 \text{ kg/s per pu}, \quad (22)$$

anchored so that rated output burns $22 \text{ MW} / (0.365 \times 49 \text{ MJ/kg}) = 1.230 \text{ kg/s}$ ($\eta_{\text{design}} = 0.365$, status ESTIMATED — see the constants dossier; cross-checked against the NAVEDTRA figure of 9,000 lb/h at 26,250 shaft hp $\Rightarrow 35.3\%$ shaft efficiency). The pre-V&V code instead fed the raw *valve stroke* into a surrogate expecting a *power fraction*, double-counting no-load fuel (defect G2) — every prior fuel total was biased by it.

5 Two-mass gas-path model (Tier C)

The LM2500 is a *two-spool* engine: the gas-generator spool (HP compressor + HP turbine, $\sim 9,500$ rpm at full power) is connected to the power-turbine/generator spool (3,600 rpm) *only by flowing gas*, not by a shaft. When the fuel valve opens, the gas generator must spool up before extra power reaches the generator — a lag invisible to single-mass models. The two-mass model:

$$2H_{hp} \dot{\omega}_{hp} = P_m^{(t)} - P_c, \quad P_c = \max[0, K_c (\omega_{hp} - \omega_{hp}^{\text{idle}})], \quad (23)$$

$$2H_{pt} \dot{\omega}_{pt} = k_b P_c - P_e^{(g)} - D_{pt} (\omega_{pt} - 1), \quad (24)$$

with $\omega_{hp}^{\text{idle}} = 4900/9500 = 0.516$ (pinned to the LM2500 Pocket Guide) and $K_c = 1/(1 - \omega_{hp}^{\text{idle}})$ normalized so full NGG speed delivers rated power. The governor’s speed input is ω_{pt} (the speed pickups are on the power-turbine shaft).

Honesty box: the linear coupling law $P_c(\omega_{hp})$ is a PLACEHOLDER. Real free-turbine power vs. gas-generator speed is strongly nonlinear ($\sim$ cubic) and depends on PT speed; the implied spool time constant $2H_{hp}/K_c \approx 0.29$ s vs. a real NGG’s 0.5–1.5 s. Tier C transient *deltas* (e.g. “the nadir is deeper than single-mass models predict”) are qualitative until this is calibrated against a recorded spool-up trace. The verification suite still holds the model to exact invariants: the steady NGG-speed-to-load map, an HP-rotor energy balance $2H_{hp} \Delta\omega_{hp} = \int (P_m^{(t)} - P_c) dt$ to 10^{-3} , and equilibrium initialization. (A former `delta_hp` “angle” state, which drifted secularly because the HP spool has no meaningful electrical angle, was removed — caught by the equilibrium test.)

6 Torsional shaft and fatigue model (Tier E)

The one *physical* shaft in the machine — the coupling between the power turbine and the generator rotor — twists elastically and has a torsional resonance. Load steps ring it; the ringing consumes fatigue life.

6.1 Dynamics, and a per-unit derivation worth reading

Two masses ($M = 2H$) joined by a torsional spring k and damper c , tracked in *differential* coordinates (θ = twist angle, ω_d = speed difference; the common-mode speed is imposed from the Tier C solution so it cannot drift):

$$\dot{\theta} = \omega_0 \omega_d, \quad \dot{\omega}_d = \frac{\tau_{pt}}{M_{pt}} + \frac{\tau_e}{M_{gen}} - \frac{T_{\text{shaft}}}{I_r}, \quad T_{\text{shaft}} = k\theta + c\omega_d, \quad I_r = \frac{M_{pt}M_{gen}}{M_{pt} + M_{gen}}. \quad (25)$$

Combining gives $\ddot{\theta} = -(\omega_0 k/I_r)\theta + \dots$, so the natural frequency is

$$\boxed{\omega_n = \sqrt{\frac{\omega_0 k}{I_r}} \implies k = \frac{\omega_n^2 I_r}{\omega_0}} \quad (26)$$

Note the factor $\omega_0 = 377$ rad/s that appears because pu speed multiplies ω_0 in the angle equation but the inertia equation is already normalized. The repository’s own history makes the point: an earlier version omitted it, silently making the “22 Hz” mode actually 427 Hz. *Per-unit derivations must be done, not assumed.* The identity (26) is now pinned by a test ($f_n = 22.0000$ Hz).

6.2 From torque history to fatigue damage

1. **Stress:** $\tau = T/W_t$ with the polar section modulus of a hollow shaft $W_t = \frac{\pi}{16} (D^4 - d^4)/D$.
2. **Detrend:** `detrend_rolling_median` separates slow operating-point shifts (low-cycle events, counted separately) from the oscillations that rainflow should count — rainflow on a non-stationary signal buries the oscillation cycles under the trend.
3. **Rainflow** (ASTM E1049 4-point method) reduces the stress history to cycles with amplitude S_a and mean S_m .

For the newcomer: what rainflow counting does

Fatigue damage is caused by closed stress–strain hysteresis loops, not by the raw wiggles of the signal. Rainflow finds those loops: reduce the signal to its turning points; then, scanning a stack of the last three turning points, whenever the middle excursion is smaller than or equal to the one that follows it, that middle excursion is a closed cycle — extract it (range = its height, mean = its midpoint) and continue. Leftover half-cycles at the end count with weight $\frac{1}{2}$. The result is a histogram of (amplitude, mean) pairs that the S–N curve can consume. Rainflow assumes a roughly stationary signal — hence the detrending step before it.

4. **Goodman correction** penalizes cycles riding on a mean stress: $S_a^{eq} = S_a/(1 - S_m/S_{u,\text{shear}})$, $S_{u,\text{shear}} = 0.6 S_u$.
5. **Basquin S–N + Miner’s rule:** $N_f = N_{\text{ref}} (S_{a,\text{ref}}/S_a^{eq})^m$ and damage $D = \sum n_i/N_{f,i}$; failure at $D = 1$.

All material and geometry numbers are PLACEHOLDER-class (typical AISI 4340, invented shaft dimensions, assumed 22 Hz mode): the model is a duty-cycle *screening* tool. Its screening conclusion stands: Load17-class demand profiles (3 s cadence) are far too slow to excite the torsional mode — shaft fatigue risk comes from faults, motor starts, and breaker events, not AI load-following.

7 Gas-plant thermodynamic models

7.1 Willans line: part-load fuel and efficiency

For dispatch purposes a gas turbine is well described by the *Willans line* — fuel flow affine in load:

$$\dot{m}_f(L) = \dot{m}_{f,\text{rated}} [b + (1 - b)L] \implies \eta(L) = \eta_{\text{fl}} \frac{L}{b + (1 - b)L}, \quad (27)$$

with no-load fuel fraction $b = 0.2$ (deliberately the same number as GGOV1’s W_{fnt}). For the LM9000 ($\eta_{\text{fl}} = 0.3952$) this gives 32.9 % at half load and 22.0 % at 20 % load — consistent with real aeroderivative behavior and with the ThermoPower surrogate’s shape. It replaced a quadratic that claimed 39.0 % at half load (defect P1; physically impossible for a machine with idle losses).

7.2 Combined cycle energy balance

The bottoming cycle recovers exhaust heat:

$$Q_{\text{HRSG}} = \dot{m}_{\text{exh}} c_p \max(T_{\text{exh}} - T_{\text{stack}}, 0), \quad P_{st} = \eta_b Q_{\text{HRSG}}, \quad \eta_b = \eta_{b,\text{nom}} \frac{T_{\text{exh}} - T_{\text{stack}}}{T_{\text{exh,nom}} - T_{\text{stack}}}. \quad (28)$$

The linear derate of η_b with exhaust temperature is a pragmatic fit form (an earlier claim that it came from PES-TR1/CIGRE-238 was wrong and is retracted in the code). For the LM9000 CC, $\eta_{b,\text{nom}}$ is *auto-tuned* so the design point hits the datasheet 72.471 MW exactly (solution: 0.288); fuel, power, and CO₂ then all remain physical with no post-hoc rescaling.

7.3 CO₂ accounting

$\dot{m}_{\text{CO}_2} = f \cdot \dot{m}_{\text{fuel}}$ with $f = 2.75$ (stoichiometric CH₄, 44.009/16.043) for generic natural gas and $f = 2.65$ for the LM9000 (back-calculated from its datasheet 492.9 kg/MWh; the EPA pipeline-gas factor converts to ≈ 2.63 kg/kg — consistent).

8 The Rowen/Hannett model (for validation only)

To compare against the 1993 field-test literature on its own terms, the Rowen (1983) simplified gas-turbine model, in Hannett & Khan’s parameterization, was implemented:

$$e = \omega_{\text{ref}} - \omega \rightarrow \underbrace{\frac{W(Xs + 1)}{Ys + Z}}_{\text{governor}} \rightarrow \text{clip} \rightarrow \underbrace{\frac{a}{bs + c}}_{\text{valve positioner}} \rightarrow \underbrace{\frac{1}{tfs + 1}}_{\text{fuel system}} \rightarrow W_f \quad (29)$$

$$W'_f = (K_3 V_{ce} + K_6) \omega \quad (K_6 = 1 - K_3), \quad \text{lag } T_{CD}, \quad f_2 = a_{f2} + b_{f2} W_f + c_{f2} (1 - \omega), \quad P_m = f_2 \omega. \quad (30)$$

Droop = Z/W . The “Typical Gas” torque law is $f_2 = 1.3 W_f - 0.299 + 0.5(1 - \omega)$, which satisfies the steady-state identity $f_2(W_f=1, \omega=1) = 1$ (the coefficients encode the same no-load-fuel physics as GGOV1’s $K_{\text{turb}}, W_{\text{fnl}}$: $b_{f2} K_3 \approx 1$, $a_{f2} + b_{f2} K_6 \approx 0$). The temperature loop is omitted: the paper states it sits at its maximum limit throughout these tests. The vendored table’s “Typical Gas” $c_{f2} = 1.5$ is treated as a transcription artifact (Rowen 1983 and every field-derived unit row use 0.5).

9 Verification results (Phase 1)

9.1 Cross-simulation against ANDES

The strongest verification available: run *the same* single-machine islanded scenario (GENCLS $H=2.8$ s, TGOV1 $R=0.04$, $T_{1,2,3}=0.15/0.3/1.5$ s, constant-power load stepped 15→18 MW) in this repository’s scipy code and in ANDES 2.0 (an independently developed, symbolically-compiled DAE simulator), and compare trajectories point-by-point.

	ANDES 2.0	scipy (this repo)
frequency nadir	59.2326 Hz	59.2326 Hz
final frequency	59.6870 Hz	59.6870 Hz
max $ \Delta\omega $ over 30 s	0.124 mHz	
max $ \Delta P_m $	8.6×10^{-5} pu (2 kW)	

Table 3: TGOV1 cross-simulation (`pixi run vv-crosscheck`). Agreement is at solver tolerance.

Reaching this agreement required (and therefore documented) three facts about ANDES that were previously assumptions: (i) a single-bus case leaves the machine’s network equations degenerate (frequency frozen) — a two-bus case is required; (ii) the comparison line must be lossless,

since a 33 kW I^2r loss shows up as a 3.4 mHz steady-state offset; (iii) ANDES’s classical machine uses the speed-voltage approximation, under which its swing equation *is* the power form (4), and its governor output enters the torque equation one-to-one — verified in the ANDES source (GEN-Base/GENCLS), not guessed. ANDES has no GGOV1 model, so GGOV1 is verified by the analytic methods below instead (a documented deviation from the plan).

9.2 Small-signal verification of GGOV1

At operating points 5, 11.5, 15, 20, and 21.9 MW (`pixi run vv-smallsignal`):

- **Equilibrium:** $\|\text{RHS}\|_\infty < 3 \times 10^{-15}$ — the initialization of §4.8 is exact.
- **Stability:** the numerically linearized 12×12 Jacobian has exactly one zero eigenvalue (the rotor angle, which has no feedback path in an islanded constant-power case — structural, expected) and all other eigenvalues with $\text{Re} \leq -0.2 \text{ s}^{-1}$.
- **Droop identity:** measured $R = 0.04000$ (error 0.000 %).
- **Isochronous:** after a load step with `rselect=0`, final $|\Delta f| = 0.0000 \text{ mHz}$.
- **Torsional:** $f_n = 22.0000 \text{ Hz}$ vs. 22 design (closed form (26)).

9.3 Invariant test suite

36 automated tests (`pixi run test`) encode, among others: the 22 MW thermal cap (temperature limiter selected at the end of an over-demand transient), the 26.4 MW transient ceiling, the initial-load guard, the D_m damping direction, K_{bc} insensitivity, fuel calibration at rated, all four `rselect` modes, TGOV1’s VMAX non-windup, the multishaft energy balance and NGG-speed/load map, Tier B/C fuel agreement, torsional frequency/damping identities, LM9000 Willans values, fuel affinity, the first-law bound, and the CC design point.

10 Validation results (Phase 3)

10.1 Hannett & Khan (1993), Table 3 protocol

The paper’s protocol (read from the paper itself; an earlier in-repo description of it was wrong): each unit starts at 50 % of generator MVA, takes a +10 % load step, droop set to 3 %; measure the time for P_m to reach 0.6 pu and the rotor-speed excursion.

model	$T_{60\%}$ (s)	$\Delta\omega_{min}$ (pu)
Hannett “typical model” (anchor)	1.140	−0.0039
Hannett “field-derived” (anchor)	2.320	−0.0076
GGOV1, PES-TR1 typical values	1.125	−0.0127
GGOV1, LM2500 overrides	0.950	−0.0098
Rowen “Typical Gas”	0.655	−0.0083

Table 4: Step-response validation. The GGOV1 implementation reproduces the literature typical-model time-to-power to **1.3 %**. The speed-excursion column depends on H , which the paper does not publish per unit; it is reported, not gated. The gap to the field-derived 2.32s is the paper’s own headline finding — typical models are systematically $\sim 40\%$ optimistic vs. real machines — and is the reason the constants dossier flags K_{pgov}/K_{igov} for vendor retuning.

10.2 Beluga 5 load-rejection reconstruction

The paper publishes two recorded traces of a real 6 MW load rejection (Figs. 4/5: rotor speed and fuel-demand signal), but not the unit’s MVA base, inertia, or droop. The reconstruction (Fig. 1):

- The turbine base is *pinned by the data*: the pre-event fuel demand $V_{ce,0} = 0.0965$ pu and the paper’s statement that V_{ce} pu equals P_m pu on the turbine base give $6/0.0965 = 62.2$ MW (Frame-7 class — consistent with the Beluga fleet).
- The event time within the digitized record is detected at $t_0 = 1.96$ s (speed flat before it; the simultaneous V_{ce} spike to 0.135 is treated as a breaker-transient recording artifact).
- Model: Rowen with the paper’s *field-derived* unit-2 dynamic constants (governor lag $Y = 3.05$ s — the “real units are slow” finding). Droop and H are fit to the features they physically control (droop $\rightarrow$ settling offset, $H \rightarrow$ peak amplitude): droop = 0.046, $H = 12$ s (degenerate with the unpublished governor lag — the fit identifies a consistent (H, Y) family, not a nameplate).
- **Untuned predictions:** V_{ce} minimum -0.0451 vs. measured -0.0504 (11 %); V_{ce} settle (29 %); peak time (23 %). RMS speed misfit 1.1×10^{-3} pu over the 14 s record.

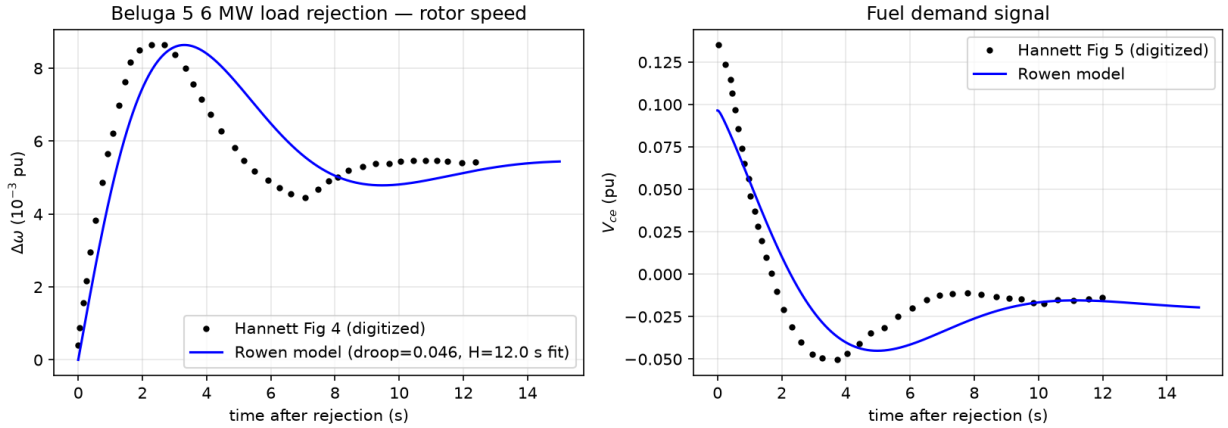


Figure 1: Beluga 5 6 MW load rejection: model (line) vs. digitized recordings (dots) from Hannett & Khan (1993) Figs. 4/5. Left: rotor speed deviation. Right: fuel demand signal V_{ce} . Droop and H fit; the V_{ce} panel is an untuned prediction.

10.3 Thermodynamic validation

First-law closure, $\dot{m}_f \text{LHV} = P + \dot{m}_{\text{exh}} c_p (T_{\text{exh}} - T_{\text{amb}}) + \text{losses}$, with c_p bracketed [1005, 1150] J/kg/K (pixi run vv-thermo):

Combined-cycle efficiencies are monotone and physical (heavy CCPP 26.1 % $\rightarrow$ 57.1 %; LM9000 CC 32.9 % $\rightarrow$ 50.5 %, design point 72.471 MW/50.49 % preserved to 0.02 %). CO_2 factors sit between stoichiometric CH_4 (2.743) and the EPA pipeline factor (≈ 2.625).

10.4 Frozen regression baselines (Phase 4)

The post-fix behavior is pinned by the automated suite so it cannot drift silently:

load	0.2	0.4	0.6	0.8	1.0
Frame GT 235 MW residual (%)	+17..+26	+3..+11	-2..+7	-4..+4	-6..+3
LM9000 SC residual (%)	+25..+32	+17..+24	+12..+19	+10..+16	+8..+14

Table 5: Unaccounted energy fraction (mechanical/generator/radiation losses plus model error). The first law is never violated (< 0 beyond the c_p bracket). The frame GT closes within the c_p uncertainty at load ≥ 0.4 . The LM9000’s larger residual quantifies its documented limitation: exhaust flow is derived from a *fixed* air-fuel ratio, so part-load exhaust energy is understated (real machines hold airflow much flatter than fuel).

LHV heat rate (kJ/kWh) at load:	0.2	0.4	0.6	0.8	1.0
Frame GT 235 MW (ThermoPower)	21,658	13,063	10,807	9,729	9,083
Heavy CCPP 338.7 MW	13,806	8,617	6,999	6,574	6,301
LM9000 simple cycle	16,397	11,842	10,324	9,565	9,109
LM9000 combined cycle	10,933	8,702	7,853	7,406	7,130

Table 6: Part-load heat rates from the post-fix models. Full-load values sit in the expected class ranges (F-class SC $\sim 9,100$; modern 1×1 CC $\sim 6,300$; aero SC $\sim 9,100$; aero CC 7,130 = the LM9000 datasheet’s 7,132 to 0.03 %).

10.5 ANDES islanded case, v2

The electrical-side case builder was upgraded (GENROU 6th-order machine with an EXST1 exciter actually attached; $S_n = P_{\text{rated}}/0.85$; governor limit at the turbine rating; and a reclose *guard*: the resync event is refused unless explicitly forced, because no sync-check relay is modeled and an out-of-phase reclosure of an island is a catastrophic event in reality). Five tests pin the ANDES interface facts numerically, including one newly discovered: ANDES converts governor power limits to the *system* base at setup ($\text{VMAX.v} = V^{\text{max}} S_n / S_{\text{sys}}$).

11 Constants: what is defensible and what is not

The dossier (`docs/constants.md`) classifies every number. The short version a reviewer should hear first:

- **Pinned:** all GGOV1 block parameters (PES-TR-1 Appendix C, transcription verified value-by-value); NGG idle/full speeds (Pocket Guide); LHV; CO₂ factors; droop convention.
- **Estimated:** $T_{\text{act}} = 0.15$ s (electronic vs. hydraulic actuator argument); $\eta_{\text{design}} = 0.365$ (literature range 35–38 %, NAVEDTRA cross-check); $S_n = 23$ MVA / $T_{\text{rate}} = 22$ MW (site convention; the Pocket Guide’s 26,250 BHP ≈ 19.6 MW *shaft* figure for the Navy variant is a documented tension); Willans $b = 0.2$.
- **Placeholder** (must not be presented as machine-specific): $H = 2.8$ s (*the* most consequential unpinned number — nadir $\propto 1/H$; publish sensitivity over $H \in [1.5, 3.5]$ s until a vendor WR² arrives); the Tier C coupling law and inertia split; the entire Tier E shaft/material/S–N set; c_p of flue gas (two values in sibling modules).

12 Limitations and recommended next steps

1. Pin H (vendor WR² or an inertia test); until then quote nadir sensitivity.

scenario	metric	pinned value
GGOV1, 11.5→13.8 MW step	nadir / final	59.450 / 59.758 Hz
TGOV1, 15→18 MW step	nadir / final	59.300 / 59.701 Hz
Two-mass, 15→18 MW step	nadir	58.772 Hz
GGOV1 at 15 MW steady	fuel	0.9290 kg/s
Load17 replay (first 10 min, ×10 MW)	$ \Delta f _{max}$ / fuel	228.3 mHz / 640.1 kg

Table 7: Regression pins in `tests/test_dynamics.py`. These supersede the historical tier-table numbers, which were produced by the pre-fix code (mixed bases, double damping, biased fuel mapping). Note the two-mass nadir is deeper than the single-mass models’ — the spool-up lag delays power delivery — but its magnitude is qualitative until the coupling law is calibrated (§5).

2. Archive the LM9000 datasheet PDF and the LM2500 gen-set heat-rate sheet (upgrades two ESTIMATED anchors to PINNED).
3. Retune K_{pgov}/K_{igov} against vendor step data — Hannett’s field-derived governors are 2–4× hotter than PES-TR1 typicals, and the 2.32 s vs. 1.14 s gap in Table 3 is exactly this effect.
4. Calibrate the Tier C spool coupling to a recorded NGG acceleration.
5. Decide droop vs. isochronous for the islanded use case (`rselect` 1 vs. 0; both now implemented and verified) — an open user decision.
6. The Tier E fatigue model needs shaft drawings and S–N data before any absolute life claim; today it is a screening tool.
7. Voltage dynamics: the scipy path holds $V = 1$ pu; studies of motor starts, faults, or reactive events must use the ANDES v2 case.

A Defect log from the pre-V&V review (abridged)

Full text in `docs/vv_log.md` Part 1. All fixed in Phase 0 and covered by tests.

- G1 (per-unit bases)** GGOV1 ran on the generator base with a hand-edited valve limit. Consequences: three inconsistent “thermal limits” (26.1, 23, 22 MW), an initial-load guard that admitted 34.7 MW, and tier-to-tier reinterpretation of V^{min} . Fix: turbine-base governor per PES-TR1 §3.3 with the single k_b conversion.
- G2 (fuel mapping)** The fuel-valve signal (where 0.2 = zero power) was fed to a surrogate expecting a power fraction — double-counting no-load fuel and biasing all fuel/CO₂ totals. Fix: native calibration §4.9.
- G3 (D_m sign)** Turbine “damping” injected energy on overspeed. Latent (default 0) but armed.
- G4 (structure)** `rselect` advertised but not implemented; TGOV1 limiter misplaced with a spurious extra pole; temperature path fed by back-derived power instead of fuel flow; reported P_e inconsistent with the P_e actually simulated.
- G5 (damping double-count)** Machine $D = 1$ and load $\alpha = 1.5$ simultaneously.
- P1 (LM9000 efficiency)** Part-load efficiency nearly flat (39.0% at half load) — replaced by the Willans line.
- P4 (citations)** A bottoming-cycle derate form mis-attributed to PES-TR1/CIGRE-238 — retracted.
- Process** An unedited LLM chat transcript lived in `docs/` as apparent reference material — quarantined with a provenance warning.

B Reproducing every number in this report

```
cd DataCenter
pixi run vv-all          # 36 tests + smoke + ANDES cross-check
                          # + small-signal + Hannett + thermo
```

Individual pieces: `pixi run test`, `vv-crosscheck`, `vv-smallsignal`, `vv-hannett`, `vv-thermo`.

References

- [1] IEEE PES Task Force on Turbine-Governor Modeling, “Dynamic Models for Turbine-Governors in Power System Studies,” Technical Report PES-TR1, Jan. 2013. (Appendix C transcription vendored in `papers/`.)
- [2] L. N. Hannett and A. Khan, “Combustion turbine dynamic model validation from tests,” *IEEE Trans. Power Systems*, vol. 8, no. 1, pp. 152–158, 1993.
- [3] W. I. Rowen, “Simplified mathematical representations of heavy-duty gas turbines,” *ASME J. Eng. Power*, 105(4), 1983.
- [4] P. Kundur, *Power System Stability and Control*, McGraw-Hill, 1994 (§11.1.4 for load damping).
- [5] S. K. Yee, J. V. Milanović, and F. M. Hughes, “Overview and comparative analysis of gas turbine models for system stability studies,” *IEEE Trans. Power Systems*, 23(1), 2008.
- [6] CIGRE Task Force 38.02.25, “Modeling of Gas Turbines and Steam Turbines in Combined-Cycle Power Plants,” Technical Brochure 238, 2003.